

Fig. 9: An actual-size, single-side PCB for the LCD extension circuit

AVR→Select AVR Programmer. Press 'Connect...' after selecting STK500 or AVRISP in the platform window and COM port (say, COM1) in the port window. Next, select Atmega2560 as 'Device', ISP as 'Programming Mode' and browse your project hex file from Project→Debug→Exe folder. Press 'Program' button on the STK500 dialogue box to burn the hex file into your microcontroller.

Converting the image file. You may convert any image file into a .raw file of 132×132 size, in 12-bit RGB colour, using Adobe Photoshop.

A standard 12-bit image encodes the two pixels in three bytes of data (24 bits) as shown in Fig. 5. In a 12-bit image, each pixel has three colour components of 4-bit value (0 to 15). So each of red, green and blue can vary from 0 to 15. Now, as a single COM port communication or any serial communication has 8-bit data and 12 bits are one-and-a-half byte, the two pixels are transferred so that 12×2 = 24-bit value is sent in three successive transfers of 8-bit value. Same is the case when storing these pixels in a file or a memory which stores a byte value only. The two pixels having 24 bits can be stored in three bytes of data.

A front-end (user-interface) program for transferring a picture file to the AVR microcontroller may be designed using a suitable software.

Using GUI to transfer the image.

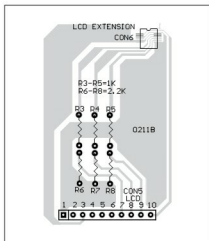


Fig. 10: Component layout for the PCB in Fig. 9

The user-interface program for transferring the picture file (.raw) from your PC to the Nokia LCD is designed in Visual Basic.NET platform. The send_picture.exe file is already included in the EFY-CD of this issue. When you click this file, you will see a GUI window as shown in Fig. 6.

In the first drop-down menu, select the appropriate com port. The image to be transferred should be saved in *.raw format. Select that image using 'Browse' option in the second pane and press 'Send' button to transfer the image to the graphical LCD. Now the image is displayed on the LCD.

Writing a text to the screen is similar to drawing a bitmap with predefined height, width and colour of characters.

Construction

An actual-size, single-side PCB for interfacing the Nokia colour LCD with the AVR microcontroller circuit (Fig. 2) is shown in Fig. 7 and its component layout in Fig. 8. An actual-size, single-side PCB for the LCD extension circuit (Fig. 4) is shown in Fig. 9 and its component layout in Fig. 10.

Mount all the components as shown in the PCB layouts. The Atmega2560 used in the prototype comes in a 100-pin, surface-mount TQFP package. By using surface-mount devices (SMDs) in your project, you don't have to drill holes in the PCB and the board size will be much smaller. But it may

be difficult to handle and solder SMDs without an SMD soldering station. However, with a normal soldering iron, tweezers and some basic knowledge, you can solder SMD components without much problem. You need good eyes, a steady hand and a soldering iron with a small, clean tip.

Make sure that the PCB is clean. You may use copper polish material to remove all kinds of oxidation and acetone (nail polish remover) to remove all kinds of contaminations on the PCB.

On the PCB, figure out the correct location for placing the SMDs. Fixate the SMD at the top right corner and the bottom left corner. Carefully solder SMD pins one by one. SMD components are very sensitive to heat, so allow your SMD to cool down after every step. For more tips on soldering SMD components, you may refer to <http://hem.passagen.se/communication/pcbasm.html> and www.engadget.com/2006/03/07/how-to-make-a-surface-mount-soldering-iron/ websites.

After soldering the components on the PCBs, insert the LCD connector extension board in the main interfacing PCB using the connector marked 'CON2'. With the LCD screen facing towards you, the leftmost pin of the berg strip is pin 1. Make sure that the orientation of the LCD connector is correct.

Now connect 9V DC source to the circuit and switch on the power supply. LED1 should glow to indicate the presence of power in the circuit. At this point, you will see a test colour band drawn by itself on the LCD screen and then waits for serial port to get 12-bit RGB image from PC at 115.2 kbps baud rate. Now run the send_picture.exe file and send the picture file (.raw) to the LCD from your computer.

EFY note. The source codes of this article (main.c and Visual Basic.NET GUI program) have been included in this month's EFY-CD and are also available on efymag.com website. ●

The author is an assistant professor in Department of Mechanical Engineering at Birla Institute of Technology, Mesra, Ranchi